

Representations

The distribution that we will most commonly use is the one-dimensional Gaussian distribution.

Definition 1 (One-Dimensional Gaussian Distribution). Given parameters $\mu \in \mathbb{R}$, $\sigma \in \mathbb{R}^+$, $\tau \in \mathbb{R}$, and $\rho \in \mathbb{R}^+$, a one-dimensional random variable $X \in \mathbb{R}$ is distributed according to a one-dimensional Gaussian distribution if the density has the form $\mathcal{N}(\cdot; \mu, \sigma^2)$ or $\mathcal{G}(\cdot; \tau, \rho)$ with

$$\mathcal{N}(x; \mu, \sigma^2) := \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right), \quad (1)$$

$$\mathcal{G}(x; \tau, \rho) := \sqrt{\frac{\rho}{2\pi}} \cdot \exp\left(-\frac{\tau^2}{2\rho}\right) \cdot \exp\left(\tau \cdot x - \rho \cdot \frac{x^2}{2}\right). \quad (2)$$

If $\mu = \tau = 0$ and $\sigma^2 = \rho = 1$ we simply write

$$\mathcal{N}(x) := \mathcal{N}(x; 0, 1), \quad (3)$$

$$\mathcal{G}(x) := \mathcal{G}(x; 0, 1). \quad (4)$$

Note that both definitions are identical when using the identities

$$\tau = \frac{\mu}{\sigma^2} \quad \text{and} \quad \rho = \sigma^{-2}, \quad \text{or} \quad (5)$$

$$\mu = \frac{\tau}{\rho} \quad \text{and} \quad \sigma^2 = \rho^{-1}. \quad (6)$$

Also note that if $X \sim \mathcal{N}(\cdot; \mu, \sigma^2)$ we know that $\mathbb{E}[X] = \mu$, $\mathbb{E}[X^2] = \mu^2 + \sigma^2$, and $\mathbb{V}[X] = \sigma^2$.

The following lemma will be useful for later proofs about the multiplication and division of Gaussian distributions.

Lemma 1 (Gaussian Density Transformation Lemma). Given parameters $\mu \in \mathbb{R}$, $\sigma \in \mathbb{R}^+$, and $a \neq 0$ we have

$$\mathcal{N}(ax; \mu, \sigma^2) = \frac{1}{|a|} \cdot \mathcal{N}\left(x; \frac{\mu}{a}, \frac{\sigma^2}{a^2}\right). \quad (7)$$

Proof. Using (1) we see that

$$\begin{aligned} \mathcal{N}(ax; \mu, \sigma^2) &= \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot \exp\left(-\frac{1}{2} \cdot \frac{(ax - \mu)^2}{\sigma^2}\right) \\ &= \frac{1}{\sqrt{2\pi} \cdot \sigma} \cdot \frac{|a|}{|a|} \cdot \exp\left(-\frac{1}{2} \cdot \frac{(a(x - a^{-1}\mu))^2}{\sigma^2}\right) \\ &= \frac{1}{|a|} \cdot \frac{1}{\sqrt{2\pi} \cdot (|a|^{-1} \cdot \sigma)} \cdot \exp\left(-\frac{1}{2} \cdot \frac{(x - a^{-1}\mu)^2}{(a^{-1} \cdot \sigma)^2}\right) \\ &= \frac{1}{|a|} \cdot \mathcal{N}\left(x; \frac{\mu}{a}, \frac{\sigma^2}{a^2}\right). \end{aligned}$$

□

Multiplication

A central theorem that allows us to derive many of our efficient factor update equations is the one-dimensional Gaussian multiplication theorem (see Figure 1 (left) for an example).

Theorem 1 (Gaussian Multiplication Theorem). *Given two Gaussian densities $\mathcal{G}(\cdot; \tau_1, \rho_1)$ and $\mathcal{G}(\cdot; \tau_2, \rho_2)$ we have*

$$\mathcal{G}(x; \tau_1, \rho_1) \cdot \mathcal{G}(x; \tau_2, \rho_2) = \mathcal{G}(x; \tau_1 + \tau_2, \rho_1 + \rho_2) \cdot \mathcal{N}(\mu_1; \mu_2, \sigma_1^2 + \sigma_2^2). \quad (8)$$

Thus, when multiplying two Gaussian densities, we obtain a non-normalized Gaussian with the precision-mean τ as the sum of the precision-means of the two factors and the precision ρ as the sum of the precisions of the two factors. The normalization constant is the density value at 0 when subtracting both means and summing both variances of the factors.

Proof. In order to prove this theorem, we show that

$$\frac{\mathcal{G}(x; \tau_1, \rho_1) \cdot \mathcal{G}(x; \tau_2, \rho_2)}{\mathcal{G}(x; \tau_1 + \tau_2, \rho_1 + \rho_2)} = \mathcal{N}(\mu_1; \mu_2, \sigma_1^2 + \sigma_2^2). \quad (9)$$

Using (2), we see that the left-hand side of (9) equals

$$\sqrt{\frac{\rho_1 \rho_2}{2\pi \cdot (\rho_1 + \rho_2)}} \cdot \exp\left(-\frac{1}{2} \cdot \left[\frac{\tau_1^2}{\rho_1} + \frac{\tau_2^2}{\rho_2} - \frac{(\tau_1 + \tau_2)^2}{\rho_1 + \rho_2}\right]\right). \quad (10)$$

At the same time, using (1), we see that the right-hand side of (9) equals

$$\sqrt{\frac{1}{2\pi \cdot (\sigma_1^2 + \sigma_2^2)}} \cdot \exp\left(-\frac{1}{2} \cdot \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2}\right). \quad (11)$$

It remains to show that (10) equals (11) using the identities (6). First, notice that

$$\sigma_1^2 + \sigma_2^2 = \rho_1^{-1} + \rho_2^{-1} = \rho_1^{-1} \rho_2^{-1} \cdot (\rho_1 + \rho_2) = \frac{\rho_1 + \rho_2}{\rho_1 \rho_2},$$

which shows the equivalence of the square-root term of (10) and (11). At the same time, we have

$$(\mu_1 - \mu_2)^2 = (\tau_1 \rho_1^{-1} - \tau_2 \rho_2^{-1})^2 = (\rho_1^{-1} \rho_2^{-1} \cdot (\tau_1 \rho_2 - \tau_2 \rho_1))^2 \Leftrightarrow \frac{(\mu_1 - \mu_2)^2}{\sigma_1^2 + \sigma_2^2} = \frac{(\tau_1 \rho_2 - \tau_2 \rho_1)^2}{\rho_1 \rho_2 \cdot (\rho_1 + \rho_2)}.$$

To complete the proof, we see that

$$\begin{aligned} \frac{\tau_1^2}{\rho_1} + \frac{\tau_2^2}{\rho_2} - \frac{(\tau_1 + \tau_2)^2}{\rho_1 + \rho_2} &= \frac{\tau_1^2 \rho_2 \cdot (\rho_1 + \rho_2) + \tau_2^2 \rho_1 \cdot (\rho_1 + \rho_2) - \rho_1 \rho_2 \cdot (\tau_1 + \tau_2)^2}{\rho_1 \rho_2 \cdot (\rho_1 + \rho_2)} \\ &= \frac{(\tau_1 \rho_2)^2 + (\tau_2 \rho_1)^2 - 2(\tau_1 \rho_2)(\tau_2 \rho_1)}{\rho_1 \rho_2 \cdot (\rho_1 + \rho_2)} \\ &= \frac{(\tau_1 \rho_2 - \tau_2 \rho_1)^2}{\rho_1 \rho_2 \cdot (\rho_1 + \rho_2)}, \end{aligned}$$

which shows the equivalence of the exponential term of (10) and (11). $\square$

Division

Using this result, it is easy to show the following division theorem (see Figure 1 (right) for an example).

Theorem 2 (Gaussian Division Theorem). *Given two Gaussian densities $\mathcal{G}(\cdot; \tau_1, \rho_1)$ and $\mathcal{G}(\cdot; \tau_2, \rho_2)$ with $\rho_1 > \rho_2$*

$$\frac{\mathcal{G}(x; \tau_1, \rho_1)}{\mathcal{G}(x; \tau_2, \rho_2)} = \frac{\mathcal{G}(x; \tau_1 - \tau_2, \rho_1 - \rho_2)}{\mathcal{N}(\mu_1; \mu_2, \sigma_2^2 - \sigma_1^2)} \cdot \left(\frac{\sigma_2^2}{\sigma_2^2 - \sigma_1^2}\right). \quad (12)$$

Proof. In order to prove this theorem, we use Theorem 1 in the following form:

$$\begin{aligned} \mathcal{G}(x; \tau_3, \rho_3) \cdot \mathcal{G}(x; \tau_2, \rho_2) &= \mathcal{G}(x; \tau_3 + \tau_2, \rho_3 + \rho_2) \cdot \mathcal{N}(\mu_3; \mu_2, \sigma_3^2 + \sigma_2^2), \\ \frac{\mathcal{G}(x; \tau_3 + \tau_2, \rho_3 + \rho_2)}{\mathcal{G}(x; \tau_2, \rho_2)} &= \frac{\mathcal{G}(x; \tau_3, \rho_3)}{\mathcal{N}(\mu_3; \mu_2, \sigma_3^2 + \sigma_2^2)}, \end{aligned}$$

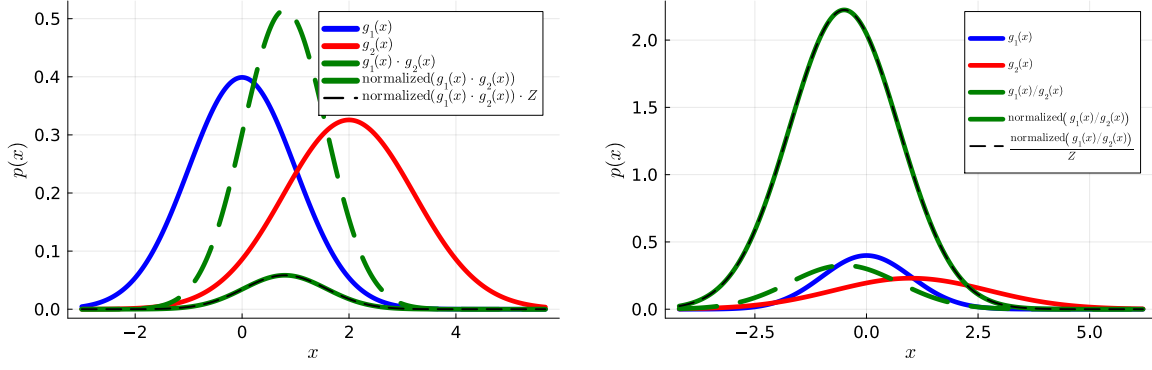


Figure 1: **(Left)** Demonstration of the Gaussian multiplication theorem with $\mathcal{N}(\cdot; 0, 1)$ (blue line) and $\mathcal{N}(\cdot; 2, \frac{3}{2})$ (red line). The point-wise product of these two densities is shown as the green line. After re-normalization to unit measure, this changes to $\mathcal{N}(\cdot; \frac{4}{5}, \frac{3}{5})$ (dashed green line). Further multiplying the normalization $\mathcal{N}(0; 2, \frac{5}{2})$ gives the same density as the point-wise product (black dashed line). **(Right)** Demonstration of the Gaussian division theorem with $\mathcal{N}(\cdot; 0, 1)$ (blue line) and $\mathcal{N}(\cdot; 1, 3)$ (red line). The point-wise division of these two densities is shown as the green line. After re-normalization to unit measure, this changes to $\mathcal{N}(\cdot; -\frac{1}{2}, \frac{3}{2})$ (dashed green line). Further dividing by the normalization $\mathcal{N}(0; 1, 2) \cdot \frac{2}{3}$ gives the same density as the point-wise division (black dashed line).

where the second line follows by division with $\mathcal{G}(x; \tau_2, \rho_2) \cdot \mathcal{N}(\mu_3; \mu_2, \sigma_3^2 + \sigma_2^2)$. All that is left is to set $\tau_1 = \tau_3 + \tau_2$ and $\rho_1 = \rho_3 + \rho_2$, substitute on the left side, and rewrite the right side in terms of μ_1, μ_2, σ_1^2 , and σ_2^2 . It is easy to see that $\tau_3 = \tau_1 - \tau_2$ and $\rho_3 = \rho_1 - \rho_2$. This implies that

$$\begin{aligned} \sigma_3^2 + \sigma_2^2 &= \frac{1}{\rho_3} + \sigma_2^2 = \frac{\sigma_1^2 \cdot \sigma_2^2}{\sigma_2^2 - \sigma_1^2} + \sigma_2^2 = \sigma_2^2 \cdot \frac{\sigma_2^2}{\sigma_2^2 - \sigma_1^2}, \\ \mu_3 - \mu_2 &= \frac{\tau_3}{\rho_3} - \mu_2 = \frac{\tau_1 - \tau_2}{\rho_1 - \rho_2} - \mu_2 = \frac{\mu_1 \sigma_2^2 - \mu_2 \sigma_1^2}{\sigma_2^2 - \sigma_1^2} - \mu_2 = (\mu_1 - \mu_2) \cdot \frac{\sigma_2^2}{\sigma_2^2 - \sigma_1^2}. \end{aligned}$$

Thus, $\mathcal{N}(\mu_3; \mu_2, \sigma_3^2 + \sigma_2^2) = \mathcal{N}(\mu_3 - \mu_2; 0, \sigma_3^2 + \sigma_2^2)$ is equivalent to

$$\mathcal{N}\left((\mu_1 - \mu_2) \cdot \frac{\sigma_2^2}{\sigma_2^2 - \sigma_1^2}; 0, \sigma_2^2 \cdot \frac{\sigma_2^2}{\sigma_2^2 - \sigma_1^2}\right) = \mathcal{N}\left(\mu_1; \mu_2, \sigma_2^2 - \sigma_1^2\right) \cdot \frac{\sigma_2^2 - \sigma_1^2}{\sigma_2^2},$$

where the final step follows from the application of Lemma 1. $\square$

Cumulative Distribution Function

An important quantity that we often use in relation to Gaussian densities is the cumulative distribution function (cdf) of the Gaussian density.

Definition 2 (Gaussian Cumulative Distribution Function). Given parameters $\mu \in \mathbb{R}, \sigma \in \mathbb{R}^+$, the cumulative distribution function $\Phi(\cdot; \mu, \sigma^2)$ of a Gaussian $\mathcal{N}(\cdot; \mu, \sigma^2)$ for any $t \in \mathbb{R}$ is defined by

$$\Phi(t; \mu, \sigma^2) := \int_{-\infty}^t \mathcal{N}(x; \mu, \sigma^2) dx, \quad (13)$$

$$\Phi(t) := \Phi(t; 0, 1). \quad (14)$$

Due to the symmetry of the Gaussian density around μ , the cumulative distribution function has some nice properties.

Theorem 3 (Gaussian Cumulative Distribution Function). Given parameters $\mu \in \mathbb{R}, \sigma \in \mathbb{R}^+$, we have for any $t \in \mathbb{R}$

$$\Phi\left(t; \mu, \sigma^2\right) = \Phi\left(\frac{t - \mu}{\sigma}\right), \quad (15)$$

$$\Phi(t) = 1 - \Phi(-t). \quad (16)$$

Proof. Using Definition 2 for the standard Gaussian density $f(x) = \mathcal{N}(x; 0, 1)$ and the transformation $g(x) = \frac{x-\mu}{\sigma}$ with $\frac{dg(x)}{dx} = \frac{1}{\sigma}$ we can use the integral substitution rule

$$\int_a^b f(g(x)) \cdot \frac{dg(x)}{dx} dx = \int_{g(b)}^{g(a)} f(u) du$$

to show that

$$\begin{aligned} \Phi(t; \mu, \sigma^2) &= \int_{-\infty}^t \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right) \cdot \frac{1}{\sigma} dx \\ &= \int_{-\infty}^{\frac{t-\mu}{\sigma}} \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{u^2}{2}\right) du \\ &= \Phi\left(\frac{t-\mu}{\sigma}\right). \end{aligned}$$

The second result follows from the symmetry of $\mathcal{N}(\cdot; 0, 1)$ around 0, that is, $\mathcal{N}(x; 0, 1) = \mathcal{N}(-x; 0, 1)$. Then

$$\begin{aligned} \Phi(t) &= \int_{-\infty}^t \mathcal{N}(x; 0, 1) dx \\ &= \int_{-\infty}^{+\infty} \mathcal{N}(x; 0, 1) dx - \int_t^{+\infty} \mathcal{N}(x; 0, 1) dx \\ &= 1 - \int_{-\infty}^{-t} \mathcal{N}(x; 0, 1) dx \\ &= 1 - \Phi(-t). \end{aligned}$$

□

Limit Gaussian Distributions: Dirac Delta

The limit case of σ^2 converging to zero and $\mu = 0$. The resulting function is called a Dirac delta and is formally defined as follows¹.

Definition 3 (Dirac Delta). The Dirac delta function $\delta(\cdot)$ is a normalized function defined as zero everywhere except at 0. Formally, it is defined by

$$\delta(x) := \lim_{\sigma^2 \rightarrow 0} \mathcal{N}(x; 0, \sigma^2). \quad (17)$$

A key property of the Dirac delta is that if a Gaussian density is convoluted with a Dirac delta, it returns the function value of the Gaussian density at zero.

Theorem 4 (Convolution of Normal with Dirac). Given a Gaussian density $\mathcal{N}(\cdot; \mu, \sigma^2)$ and a Dirac delta $\delta(\cdot)$ we know that

$$\int_{-\infty}^{+\infty} \delta(x) \mathcal{N}(x; \mu, \sigma^2) dx = \mathcal{N}(0; \mu, \sigma^2). \quad (18)$$

Proof. Using both Definition 3 and Theorem 1 we see that

$$\begin{aligned} \int_{-\infty}^{+\infty} \delta(x) \mathcal{N}(x; \mu, \sigma^2) dx &= \int_{-\infty}^{+\infty} \lim_{s^2 \rightarrow 0} \mathcal{N}(x; 0, s^2) \mathcal{N}(x; \mu, \sigma^2) dx \\ &= \lim_{s^2 \rightarrow 0} \int_{-\infty}^{+\infty} \mathcal{G}(x; \sigma^{-2}\mu, s^{-2} + \sigma^{-2}) \cdot \mathcal{N}(0; \mu, s^2 + \sigma^2) dx \\ &= \lim_{s^2 \rightarrow 0} \mathcal{N}(0; \mu, s^2 + \sigma^2) \\ &= \mathcal{N}(0; \mu, \sigma^2). \end{aligned}$$

□

¹Note that a $\mu \neq 0$ can be realized by subtracting it from the argument of $\delta(\cdot)$ because $\mathcal{N}(x; \mu, \sigma^2) = \mathcal{N}(x - \mu; 0, \sigma^2)$.

Multiplication	$\delta(\cdot - \mu_2)$	$\mathcal{G}(\cdot; \tau_2, \rho_2)$	$\mathcal{U}(\cdot)$
$\delta(\cdot - \mu_1)$	$\delta(\cdot - \mu_2)$ if $\mu_1 = \mu_2$	$\delta(\cdot - \mu_1)$	$\delta(\cdot - \mu_1)$
$\mathcal{G}(\cdot; \tau_1, \rho_1)$	$\delta(\cdot - \mu_2)$	$\mathcal{G}(\cdot; \tau_1 + \tau_2, \rho_1 + \rho_2)$	$\mathcal{G}(\cdot; \tau_1, \rho_1)$
$\mathcal{U}(\cdot)$	$\delta(\cdot - \mu_2)$	$\mathcal{G}(\cdot; \tau_2, \rho_2)$	$\mathcal{U}(\cdot)$

Table 1: Multiplication of Normal-distributed variables (without normalization constant) for the extreme cases of point-mass, $\delta(\cdot - \mu)$, density $\mathcal{G}(\cdot; \tau, \rho)$, and uniform $\mathcal{U}(\cdot)$.

Division	$\delta(\cdot - \mu_2)$	$\mathcal{G}(\cdot; \tau_2, \rho_2)$	$\mathcal{U}(\cdot)$
$\delta(\cdot - \mu_1)$	$\mathcal{U}(\cdot)$ if $\mu_1 = \mu_2$	$\delta(\cdot - \mu_1)$	$\delta(\cdot - \mu_1)$
$\mathcal{G}(\cdot; \tau_1, \rho_1)$	-	$\mathcal{G}(\cdot; \tau_1 - \tau_2, \rho_1 - \rho_2)$	$\mathcal{G}(\cdot; \tau_1, \rho_1)$
$\mathcal{U}(\cdot)$	-	-	$\mathcal{U}(\cdot)$

Table 2: Division of Normal-distributed variables (without normalization constant) for the extreme cases of point-mass, $\delta(\cdot - \mu)$, density $\mathcal{G}(\cdot; \tau, \rho)$, and uniform $\mathcal{U}(\cdot)$.

Limit Gaussian Distributions: Gaussian Uniform

The limit case of σ^2 converging to $+\infty$ and $\mu = 0$. The resulting function is effectively a constant function and is formally defined as follows.

Definition 4 (Gaussian Uniform). The Gaussian uniform function $\mathcal{U}(\cdot)$ is a normalized function defined as constant everywhere. Formally, it is defined by

$$\mathcal{U}(x) := \lim_{\sigma^2 \rightarrow +\infty} \mathcal{N}(x; 0, \sigma^2). \quad (19)$$

A key property of the Gaussian uniform is that if a Gaussian density is multiplied by a Gaussian uniform and renormalized, it returns the original Gaussian density.

Theorem 5 (Multiplication of Normal with Gaussian Uniform). Given a Gaussian density $\mathcal{N}(\cdot; \mu, \sigma^2)$ and a Gaussian uniform $\mathcal{U}(\cdot)$ we know for any $x \in \mathbb{R}$ that

$$\frac{\mathcal{U}(x) \mathcal{N}(x; \mu, \sigma^2)}{\int_{-\infty}^{+\infty} \mathcal{U}(\tilde{x}) \mathcal{N}(\tilde{x}; \mu, \sigma^2) d\tilde{x}} = \mathcal{N}(x; \mu, \sigma^2). \quad (20)$$

Proof. First note that for any $s^2 > 0$ and using Theorem 1 we have

$$\begin{aligned} \mathcal{N}(x; 0, s^2) \mathcal{N}(x; \mu, \sigma^2) &= \mathcal{G}(x; \sigma^{-2}\mu, s^{-2} + \sigma^{-2}) \cdot \mathcal{N}(0; \mu, s^2 + \sigma^2) \\ &= \mathcal{N}\left(x; \mu \cdot \frac{s^2}{s^2 + \sigma^2}, \sigma^2 \cdot \frac{s^2}{s^2 + \sigma^2}\right) \cdot \mathcal{N}(0; \mu, s^2 + \sigma^2) \\ &= \mathcal{N}\left(x; \mu \cdot \frac{1}{1 + \frac{\sigma^2}{s^2}}, \sigma^2 \cdot \frac{1}{1 + \frac{\sigma^2}{s^2}}\right) \cdot \mathcal{N}(0; \mu, s^2 + \sigma^2). \end{aligned}$$

Hence, $\int_{-\infty}^{+\infty} \mathcal{N}(\tilde{x}; 0, s^2) \mathcal{N}(\tilde{x}; \mu, \sigma^2) d\tilde{x} = \mathcal{N}(0; \mu, s^2 + \sigma^2)$ for any $s > 0$ and, using Definition 4, we see that

$$\begin{aligned} \frac{\mathcal{U}(x) \mathcal{N}(x; \mu, \sigma^2)}{\int_{-\infty}^{+\infty} \mathcal{U}(\tilde{x}) \mathcal{N}(\tilde{x}; \mu, \sigma^2) d\tilde{x}} &= \lim_{s^2 \rightarrow +\infty} \frac{\mathcal{N}(x; 0, s^2) \mathcal{N}(x; \mu, \sigma^2)}{\int_{-\infty}^{+\infty} \mathcal{N}(\tilde{x}; 0, s^2) \mathcal{N}(\tilde{x}; \mu, \sigma^2) d\tilde{x}} \\ &= \lim_{s^2 \rightarrow +\infty} \mathcal{N}\left(x; \mu \cdot \frac{1}{1 + \frac{\sigma^2}{s^2}}, \sigma^2 \cdot \frac{1}{1 + \frac{\sigma^2}{s^2}}\right) \\ &= \mathcal{N}(x; \mu, \sigma^2). \end{aligned}$$

□

In Tables 1 and 2 we have detailed the result of normalized multiplication and division using both the Dirac delta and the Gaussian uniform function as possible factors.